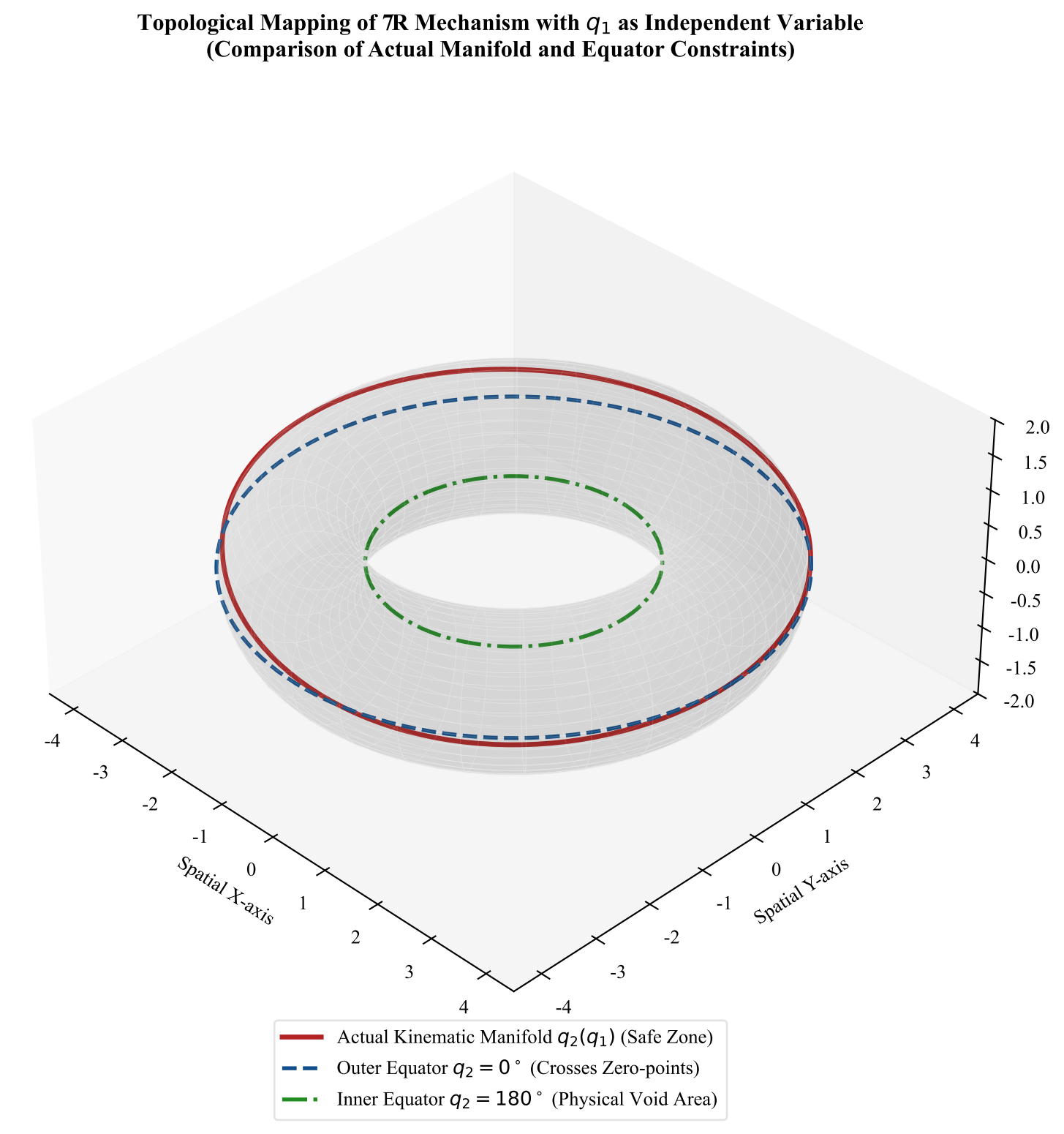

Topological Mapping of 7R Mechanism with q_1 as Independent Variable
(Comparison of Actual Manifold and Equator Constraints)
Spatial X-axis
Spatial Y-axis
-4
-3
-2
-1
0
1
2
3
4
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
Actual Kinematic Manifold $q_2(q_1)$ (Safe Zone)
Outer Equator $q_2 = 0^\circ$ (Crosses Zero-points)
Inner Equator $q_2 = 180^\circ$ (Physical Void Area)